

## SAFETY DATA SHEET

Creation Date 07-March-2012

Revision Date 29-March-2024

Revision Number 4

### 1. Identification

**Product Name** Trimethylacetic acid

**Cat No. :** L02557

**CAS-No** 75-98-9

**Synonyms** Pivalic acid; 2,2-Dimethylpropanoic acid

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                   |            |
|-----------------------------------|------------|
| Acute oral toxicity               | Category 4 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 2 |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Harmful if swallowed  
Causes skin irritation  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell

IF ON SKIN: Wash with plenty of soap and water

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Rinse mouth

If skin irritation occurs: Get medical advice/attention

If eye irritation persists: Get medical advice/attention

Take off contaminated clothing and wash it before reuse

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                     | CAS-No  | Weight % |
|-------------------------------|---------|----------|
| Propanoic acid, 2,2-dimethyl- | 75-98-9 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                                                                                                                                                                                                  |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                         |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                       |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                            |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                   |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                   |

### 5. Fire-fighting measures

|                                     |                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | 63 °C / 145.4 °F         |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Combustible material. Containers may explode when heated.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                             |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Remove all sources of ignition.                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid dust formation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition.         |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Bases. Reducing Agent. |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                                                                                                                                                                                                                                                             |
| <b>Engineering Measures</b>          | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.<br>Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |
| <b>Personal protective equipment</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Eye Protection**  
**Hand Protection**Goggles  
Protective gloves

| Glove material                                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|---------------------------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Natural rubber<br>Butyl rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                                            |
|-----------------------------------------------|--------------------------------------------|
| <b>Physical State</b>                         | Solid                                      |
| <b>Appearance</b>                             | White                                      |
| <b>Odor</b>                                   | pungent                                    |
| <b>Odor Threshold</b>                         | No information available                   |
| <b>pH</b>                                     | No information available                   |
| <b>Melting Point/Range</b>                    | 33 - 35 °C / 91.4 - 95 °F                  |
| <b>Boiling Point/Range</b>                    | 163 - 164 °C / 325.4 - 327.2 °F @ 760 mmHg |
| <b>Flash Point</b>                            | 63 °C / 145.4 °F                           |
| <b>Evaporation Rate</b>                       | Not applicable                             |
| <b>Flammability (solid,gas)</b>               | No information available                   |
| <b>Flammability or explosive limits</b>       |                                            |
| Upper                                         | No data available                          |
| Lower                                         | No data available                          |
| <b>Vapor Pressure</b>                         | 9.75 mmHg @ 60 °C                          |
| <b>Vapor Density</b>                          | Not applicable                             |
| <b>Specific Gravity</b>                       | 0.889                                      |
| <b>Solubility</b>                             | Soluble                                    |
| <b>Partition coefficient; n-octanol/water</b> | No data available                          |
| <b>Autoignition Temperature</b>               | No information available                   |
| <b>Decomposition Temperature</b>              | No information available                   |
| <b>Viscosity</b>                              | Not applicable                             |
| <b>Molecular Formula</b>                      | C5 H10 O2                                  |
| <b>Molecular Weight</b>                       | 102.13                                     |

## 10. Stability and reactivity

|                                         |                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                                                        |
| <b>Stability</b>                        | Moisture sensitive. Stable under normal conditions.                                                                                               |
| <b>Conditions to Avoid</b>              | Incompatible products. Exposure to moist air or water. Temperatures above 50°C. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Bases, Reducing Agent                                                                                                                             |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                                                                           |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                          |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                     | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation             |
|-------------------------------|--------------------------|------------------------------|-----------------------------|
| Propanoic acid, 2,2-dimethyl- | LD50 = 900 mg/kg ( Rat ) | LD50 = 3160 mg/kg ( Rabbit ) | LC50 > 5.3 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Causes burns by all exposure routes                                                      |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                     | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------|---------|------------|------------|------------|------------|------------|
| Propanoic acid, 2,2-dimethyl- | 75-98-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                                                                                                  |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                  |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                  |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                                                                                                 |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                 |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Do not flush into surface water or sanitary sewer system.

| Component                     | Freshwater Algae                                          | Freshwater Fish                                       | Microtox   | Water Flea |
|-------------------------------|-----------------------------------------------------------|-------------------------------------------------------|------------|------------|
| Propanoic acid, 2,2-dimethyl- | EC50: = 75 mg/L, 96h<br>(Pseudokirchneriella subcapitata) | LC50: > 300 mg/L, 96h static<br>(Oncorhynchus mykiss) | Not listed | Not listed |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component                     | log Pow |
|-------------------------------|---------|
| Propanoic acid, 2,2-dimethyl- | 1.8     |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

### 15. Regulatory information

#### International Inventories

| Component                     | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Propanoic acid, 2,2-dimethyl- | 75-98-9 | X   | -    | X    | ACTIVE                                        | 200-922-5 | -      | -   |

| Component                     | CAS-No  | IECSC | KECL   | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------------|---------|-------|--------|------|------|------|------|-------|-------|
| Propanoic acid, 2,2-dimethyl- | 75-98-9 | X     | 97-3-5 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                     | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Propanoic acid, 2,2-dimethyl- | Part 4 Substance                                     |                                                                            |                                           |

**Legend** NPRI - National Pollutant Release Inventory

## Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                     | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Propanoic acid, 2,2-dimethyl- | 75-98-9 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                     | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Propanoic acid, 2,2-dimethyl- | 75-98-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 07-March-2012  
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**Print Date** 29-March-2024  
**Revision Summary** New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**